

# PARTHA SARATHY S

## Freelancer

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| [Instagram](#)

## PROFESSIONAL SUMMARY

Enthusiastic and self-motivated aspiring Backend Developer with strong skills in JavaScript, TypeScript, Python, and backend frameworks like Node.js and Express.js. Experienced in developing AI-powered IoT projects and smart automation solutions through academic and personal projects. Quick learner with a passion for solving problems, building scalable applications, and continuously upgrading technical knowledge.

## SKILLS

|              |             |
|--------------|-------------|
| • React      | Beginner    |
| • Express Js | Experienced |
| • Python     | Experienced |
| • Mysql      | Experienced |
| • Java       | Experienced |
| • Node js    | Experienced |
| • Javascript | Experienced |
| • Typescript | Skillful    |

## WORK EXPERIENCE

### Student-Intern

Salem steel plant , SalemSeptember 2023 - September 2023

- Observed and learned about steel production processes, including casting, rolling, and quality checks.
- Acquired practical knowledge of industrial safety protocols, manufacturing workflows, and quality assurance methods.
- Participated in team discussions to identify process bottlenecks and potential efficiency improvements.
- Contributed to a project focused on optimizing material usage, delivering a 5% reduction in production waste.
- Documented findings and prepared reports for review by senior engineers, receiving positive feedback for detailed insights.

### Python Student-intern

Next GenSolution , SalemApril 2024 - May 2024

- Developed Python scripts for automating workflows and processing complex datasets, improving task efficiency by 30 percent.
- Assisted in debugging and optimizing existing Python codebases to reduce runtime by an average of 15 percent.
- Collaborated with cross-functional teams to integrate Python-based solutions into larger software projects.
- Implemented data validation and error-handling mechanisms to enhance script reliability and reduce downtime.
- Researched and applied advanced Python libraries such as Pandas and NumPy to streamline data analysis and reporting tasks.

## Full stack developer

Freelancer , Salem

August 2024 - Present

- Developed scalable and responsive web applications utilizing modern frameworks like React and Node.js, ensuring optimal user experience and performance.
- Collaborated with cross-functional teams to gather requirements, design solutions, and implement features, leading to a 30% increase in client satisfaction.
- Streamlined backend processes through RESTful API development and database optimization using MongoDB and MYSQL, improving application speed by 25%..

## EDUCATION

### Higher Secondary

Bala Bharathi Matriculation Higher Secondary School , Salem

Percentage: 85.3

June 2021 - April 2023

### BE EEE

Karpagam college of engineering , Coimbatore, Tamil Nadu

CGPA: 8.6

September 2023 - Present

## PROJECTS

### Smart Billing System (Freelancing — Electronics Shop)

<https://github.com/partha0025/Billing-System>

- Developed a custom Smart Billing and Inventory Management System for an electronics shop using Python and MySQL to streamline billing operations and inventory tracking.
- Implemented user-friendly interface and automated invoice generation process, reducing manual entry errors by 35 percent.
- Designed robust database schema and optimized queries to improve system response time by 40 percent.
- Enabled real-time inventory updates and detailed sales reporting, increasing operational transparency and aiding in strategic business decisions.
- Delivered the project ahead of schedule, earning 100 percent customer satisfaction.

### Data Structure library(partha\_ds)

[https://github.com/partha0025/Data\\_structure\\_libaray](https://github.com/partha0025/Data_structure_libaray)

- Designed and developed a robust custom data structure library in JavaScript to optimize algorithm efficiency and streamline complex application logic.
- Implemented key data structures including stacks, queues, linked lists, and binary trees, leveraging object-oriented programming principles.
- Enhanced computational performance by up to 30 percent through well-optimized algorithms and efficient memory management.
- Applied rigorous testing methodologies using Jest to ensure library reliability and reduced bugs by 25 percent.
- Documented all library functions with detailed comments and examples, improving developer onboarding and adoption rates.

### smart solar powered LED system

<https://partha0025.github.io/solarDashboard/>

- Designed and implemented a smart solar-powered LED system that optimized energy usage and reduced operational costs by 15 percent.
- Developed a responsive web application using JavaScript and React.js to display real-time solar power generation and energy consumption metrics.
- Integrated IoT devices and sensors, leveraging Python and AWS IoT Core for seamless communication and data processing.
- Utilized machine learning algorithms to predict energy usage trends, achieving 95 percent accuracy in forecasts.

- Streamlined data visualization with dynamic charts using Chart.js, enhancing stakeholder decision-making efficiency.

## Digital logic circuits simulation using java

<https://github.com/partha0025/Digital-logic-circuit-simulation>

- Designed and developed a Java-based console application to simulate basic digital logic circuits including AND, OR, and NOT gates.
- Implemented object-oriented programming principles to create modular and reusable classes for circuit components.
- Optimized circuit simulation algorithms to achieve 20 percent faster processing times compared to initial prototypes.
- Conducted thorough unit testing and debugging to ensure application accuracy and reliability.
- Delivered a functional and efficient tool that enhanced learning and experimentation for over 200 users.

## Eco Lift-ICA Web Application (IEEE Hackathon)

<https://partha0025.github.io/RENEWABLE-ENERGY-INTEGRATED-LAT-PULL-DOWN-MACHINE/>

- Designed and implemented the Eco Lift-ICA web application during the IEEE Hackathon to promote eco-friendly transportation alternatives using innovative features.
- Utilized technologies such as React.js for front-end development, Node.js for back-end integration, and MongoDB for data storage.
- Optimized application performance and user experience through responsive design and seamless API integration.
- Achieved recognition for technical excellence and creativity, contributing to hackathon success.

## LANGUAGES

- |           |                   |
|-----------|-------------------|
| • Tamil   | Native Speaker    |
| • English | Very Good Command |

## HOBBIES

Photography  
Coding  
Exploring Techs  
Sports  
Movies